

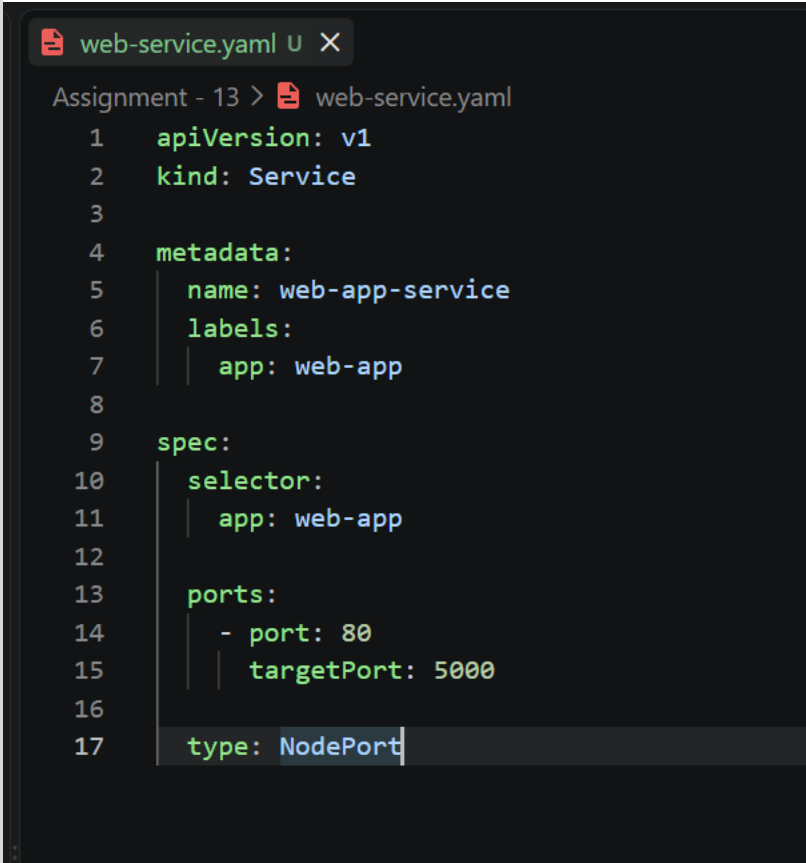


# SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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## Department of Computer Science and Engineering DevOps Lab - L3

| Lab Assignment No: 13    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of Student          | Aryan Sheladia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| PRN                      | 23070122202                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Batch                    | 2023 - 2027                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Class                    | CSE – B1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Academic Year & Semester | 2026 - 2027, 7th Semester                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Title of Assignment      | Kubernetes Service & Networking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Implementation & Theory  | <div><pre>web-service.yaml U X Assignment - 13 &gt; web-service.yaml 1  apiVersion: v1 2  kind: Service 3 4  metadata: 5    name: web-app-service 6    labels: 7      app: web-app 8 9  spec: 10   selector: 11     app: web-app 12 13   ports: 14     - port: 80 15       targetPort: 5000 16 17   type: NodePort</pre></div> <p><b>Service YAML Created:</b> A Kubernetes Service manifest named web-service.yaml was created. The Service selects the Flask application Pods using the label app: web-app, exposes port 80 internally, and forwards requests to port 5000 where the Flask application is running.</p> |

The NodePort type exposes the application outside the Kubernetes cluster.

```
PS E:\DevOps_SEM_7\Assignment - 13> kubectl apply -f web-service.yaml
service/web-app-service created
PS E:\DevOps_SEM_7\Assignment - 13> kubectl get service
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1     <none>         443/TCP          50m
web-app-service      NodePort    10.111.86.3   <none>         80:30668/TCP     5s
PS E:\DevOps_SEM_7\Assignment - 13>
```

**Service Created and Verified:** The Kubernetes Service was successfully created using `kubectl apply -f web-service.yaml`. The `kubectl get service` command confirmed that `web-app-service` is running as a NodePort Service, with Kubernetes assigning an external high-numbered port that maps to the internal Service port 80.

```
PS E:\DevOps_SEM_7\Assignment - 13> minikube service web-app-service
```

| NAMESPACE | NAME            | TARGET PORT | URL                       |
|-----------|-----------------|-------------|---------------------------|
| default   | web-app-service | 80          | http://192.168.49.2:30668 |

Starting tunnel for service web-app-service.

| NAMESPACE | NAME            | TARGET PORT | URL                    |
|-----------|-----------------|-------------|------------------------|
| default   | web-app-service |             | http://127.0.0.1:62172 |

← → ↻ ⓘ 127.0.0.1:62172

# Hello, Docker!

## Request Log

Request received at 2026-09-03 22:10:15.998488

**Application Accessed Through Kubernetes Service:** The Flask application was accessed using `minikube service web-app-service`. The successful “Hello, Docker!” response confirms that external traffic was correctly routed through the NodePort Service to the Flask application Pods running in the Kubernetes cluster.